

Name: _____

Date: _____

MULTIPLICATION, (SHORT MULTIPLICATION)

LO: I can multiply a multi-digit number by a single digit using short multiplication.

What is short multiplication?

Short multiplication is the column method for multi-digit $\times$ **SINGLE**-digit. Multiply each digit, carry tens to the next column.

When to use: multiplying by 2-9. For 10+, use long multiplication.

Examples

●	$234 \times 6 = 1404$	with carrying
●	Carry tens to next column	left-shift
●	$47 \times 8 = 376$	2-digit
●	Forget to add the carry	answer wrong
●	Multiply right to left	actually right-to-left IS correct!

Key idea

Multiply each digit (right to left). If the result is 10+, write the units and CARRY the tens to the next column.

$$\textcircled{47} \times \textcircled{8} = \textcircled{376}$$

$8 \times 7 = 56$, write 6 carry 5; then $8 \times 4 + 5 = 37$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - no carry

Calculate 32×3

$$3 \times 2 = 6 \text{ (units)}$$

$$3 \times 3 = 9 \text{ (tens)}$$

Read off: 96

$$= 96$$

No carrying needed because each multiplication is < 10 .

Example 2 - single carry

Calculate 47×8

$$8 \times 7 = 56 \text{ (write 6, carry 5)}$$

$$8 \times 4 = 32, +5 = 37$$

Read off: 376

$$= 376$$

Always work right to left, carrying any tens.

Example 3 - 3-digit number

Calculate 234×6

$$6 \times 4 = 24 \text{ (write 4, carry 2)}$$

$$6 \times 3 = 18, +2 = 20 \text{ (write 0, carry 2)}$$

$$6 \times 2 = 12, +2 = 14$$

$$= 1404$$

Multiple carries: each one adds to the next column.

Example 4 - with zero

Calculate 305×7

$$7 \times 5 = 35 \text{ (write 5, carry 3)}$$

$$7 \times 0 = 0, +3 = 3$$

$$7 \times 3 = 21$$

$$= 2135$$

Zeros multiply to give 0 — but carries still apply.

YOUR TURN – PRACTICE (deliberate practice)

1. Single-digit results

No carrying needed.

- a) 13×2
- b) 23×3
- c) 31×3
- d) 412×2

2. With carrying

Carry the tens to the next column.

- a) 47×8
- b) 56×7
- c) 83×6
- d) 94×5

3. 3-digit by 1-digit

Same method, one extra column.

- a) 234×6
- b) 457×4
- c) 623×7
- d) 856×9

4. With zero in number

Don't forget: $0 \times n = 0$, plus any carry.

- a) 305×7
- b) 402×8
- c) 709×6
- d) 5006×4

5. Mixed

Apply short multiplication.

- a) 78×6
- b) 239×5
- c) 517×8
- d) 1234×7
- e) 999×9

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Multiply right to left ☐

Correct answer: _____

Reason: _____

b) Carries are added to the NEXT column's product ☐

Correct answer: _____

Reason: _____

c) $47 \times 8 = 326$ (forget carry) ☐

Correct answer: _____

Reason: _____

d) $0 \times n = 0$, ignore the carry ☐

Correct answer: _____

Reason: _____

e) Short multiplication only works for single-digit multipliers ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A book has **247 pages**. A library has **9 copies**. How many pages in total across all copies? Use short multiplication.

Expression: _____

Simplified: _____

b) A factory makes **1234 widgets a day, 7 days a week**. How many widgets per week? How many in 4 weeks? Show your working clearly.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

